

1 Solution — GIAM 3.5

Problem 1

1.1 Problem

Prove that if n is an odd number then $n^4 \equiv 1 \pmod{16}$.

1.2 Answer

Show first that $n^2 \equiv 1 \pmod{8}$, then square.

1.3 A Lemma

Lemma. If n is odd then $n^2 \equiv 1 \pmod{8}$.

Proof. Since n is odd, $n = 2m + 1$ for some integer m . Then

$$n^2 = 4m^2 + 4m + 1 = 4m(m + 1) + 1.$$

Now m and $m + 1$ are consecutive integers, so one of them is even and their product is even: $m(m + 1) = 2j$ for some integer j . Hence

$$n^2 = 8j + 1 \equiv 1 \pmod{8}. \quad \blacksquare$$

1.4 Proof

Theorem. If n is odd then $n^4 \equiv 1 \pmod{16}$.

Proof. Let n be odd. By the Lemma there is an integer j with $n^2 = 8j + 1$. Squaring,

$$n^4 = (n^2)^2 = (8j + 1)^2 = 64j^2 + 16j + 1 = 16(4j^2 + j) + 1.$$

Since $4j^2 + j$ is an integer, $n^4 \equiv 1 \pmod{16}$. ■

Verified for every odd n up to 20 001.

1.5 The Same Result by Cases

The section's method gives an alternative route to the Lemma, replacing the “consecutive integers” observation with a two-case split. Every odd integer is of the form $4k + 1$ or $4k + 3$.

Case 1: $n = 4k + 1$. Then $n^2 = 16k^2 + 8k + 1 = 8(2k^2 + k) + 1 \equiv 1 \pmod{8}$.

Case 2: $n = 4k + 3$. Then $n^2 = 16k^2 + 24k + 9 = 8(2k^2 + 3k + 1) + 1 \equiv 1 \pmod{8}$.

The cases are exhaustive, so $n^2 \equiv 1 \pmod{8}$ for every odd n , and the theorem follows as before.

One could also case on all eight odd residues mod 16 directly, but that is eight computations rather than two — the point of choosing the right modulus to split on.

1.6 The Picture

The diagram collects the two-step proof, the case version, and a check of all eight odd residues, together with the two facts that pin the statement down: 16 cannot be enlarged and the exponent 4 cannot be reduced.

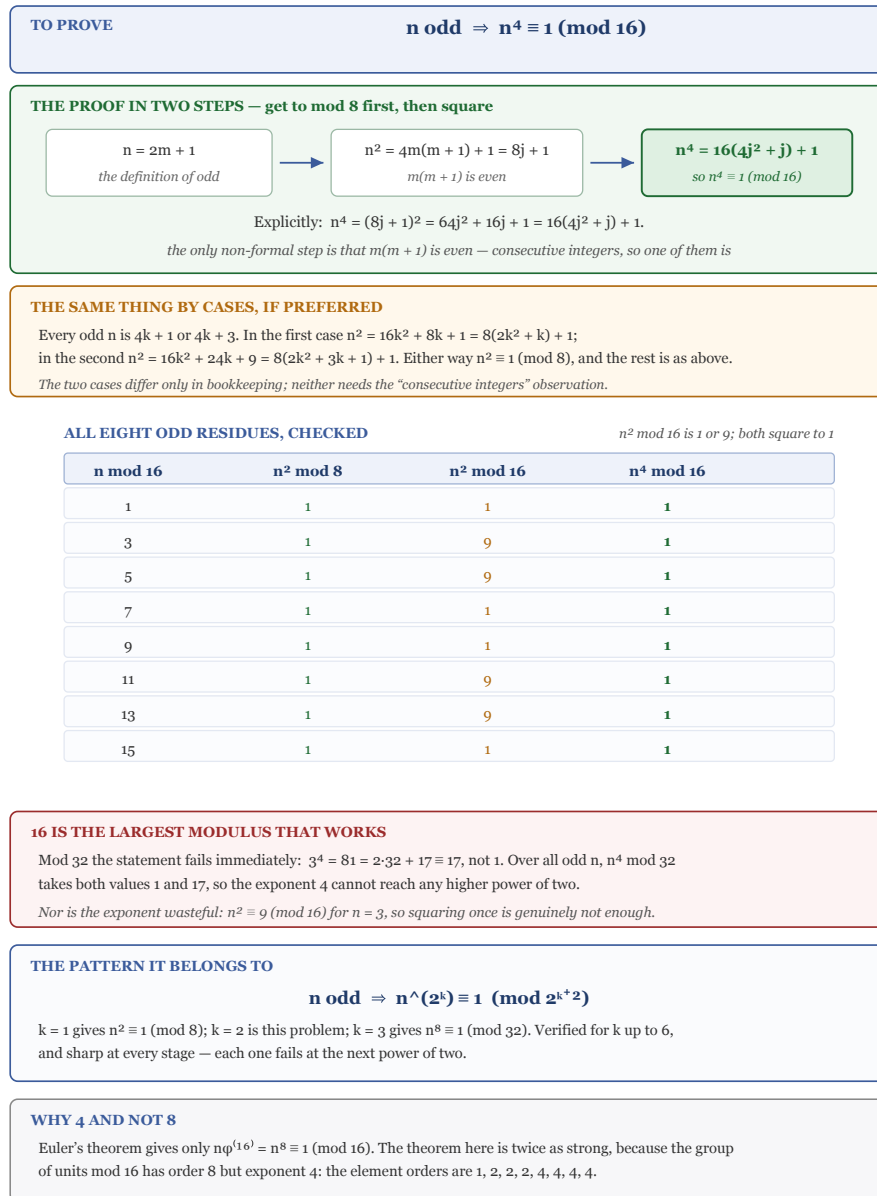


Figure 1.1: A diagram in six parts. The first states what is to be proved: n odd implies n to the fourth is congruent to 1 mod 16. The second gives the proof in two steps — get to mod 8 first, then square — as three boxes: n equals $2m$ plus 1, the definition of odd; then n squared equals $4m$ times m plus 1, plus 1, which equals $8j$ plus 1 because m times m plus 1 is even; then n to the fourth equals 16 times the quantity $4j$ squared plus j , plus 1, so n to the fourth is congruent to 1 mod 16. Explicitly, n to the fourth equals $8j$ plus 1 all squared, which is $64j$ squared plus $16j$ plus 1, which is 16 times $4j$ squared plus j , plus 1. The only non-formal step is that m times m plus 1 is even, since consecutive integers include an even one. The third band gives the same thing by cases: every odd n is $4k$ plus 1 or $4k$ plus 3; in the first case n squared is $16k$ squared plus $8k$ plus 1, which is 8 times $2k$ squared plus k , plus 1; in the second n squared is $16k$ squared plus $24k$ plus 9, which is 8 times $2k$

squared plus $3k$ plus 1, plus 1. Either way n squared is congruent to 1 mod 8. The centre is a table of all eight odd residues mod 16, listing for each of 1, 3, 5, 7, 9, 11, 13, 15 the values of n squared mod 8, n squared mod 16, and n to the fourth mod 16. The n squared mod 8 column is 1 throughout; the n squared mod 16 column alternates between 1 and 9; the n to the fourth mod 16 column is 1 throughout. A band below shows 16 is the largest modulus that works: mod 32 the statement fails immediately, since 3 to the fourth is 81, which is 2 times 32 plus 17, so congruent to 17 rather than 1; over all odd n , n to the fourth mod 32 takes both values 1 and 17. Nor is the exponent wasteful, since n squared is congruent to 9 mod 16 when n is 3, so squaring once is genuinely not enough. The next band gives the pattern it belongs to: n odd implies n to the power 2 to the k is congruent to 1 mod 2 to the k plus 2. Taking k equal to 1 gives n squared congruent to 1 mod 8; k equal to 2 is this problem; k equal to 3 gives n to the eighth congruent to 1 mod 32. This was verified for k up to 6 and is sharp at every stage, each failing at the next power of two. The last band explains why 4 and not 8: Euler's theorem gives only n to the phi of 16, that is n to the eighth, congruent to 1 mod 16, so the theorem here is twice as strong, because the group of units mod 16 has order 8 but exponent 4, its element orders being 1, 2, 2, 2, 4, 4, 4, 4.

1.7 Remark — Why Reach mod 8 First

The natural first attempt is to expand $(2m + 1)^4$ directly:

$$n^4 = 16m^4 + 32m^3 + 24m^2 + 8m + 1 = 16(m^4 + 2m^3) + 8m(3m + 1) + 1.$$

Everything reduces mod 16 except $8m(3m + 1)$, and finishing requires observing that $m(3m + 1)$ is always even — a small case argument of its own. So the direct expansion works but leaves a loose end at exactly the same place.

Squaring twice avoids the mess. The intermediate statement $n^2 \equiv 1 \pmod{8}$ is genuinely easier than the target, and once it is in hand the second squaring is pure algebra: $(8j + 1)^2$ has a $64j^2$ term and a $16j$ term, both of which vanish mod 16.

The general shape is worth noting: **squaring a residue $\equiv 1 \pmod{2^k}$ produces one $\equiv 1 \pmod{2^{k+1}}$** , since $(2^k j + 1)^2 = 2^{2k} j^2 + 2^{k+1} j + 1$. Each squaring buys one more power of two.

1.8 Remark — The Modulus Is Sharp

Sixteen cannot be improved. Mod **32**,

$$3^4 = 81 = 2 \cdot 32 + 17 \equiv 17 \pmod{32},$$

so the statement fails at the very first non-trivial odd number. Sweeping all odd n , the value of $n^4 \bmod 32$ takes both **1** and **17** — never settling.

Nor is the fourth power extravagant. Squaring only once is not enough: odd squares mod **16** take the two values $\{1, 9\}$, with $3^2 = 9$. So $n^2 \equiv 1 \pmod{16}$ is false, and the exponent **4** is the smallest that works for this modulus.

Both bounds are visible in the table: the $n^2 \bmod 16$ column alternates between **1** and **9**, while the $n^4 \bmod 16$ column is constantly **1**.

1.9 Remark — The Pattern This Belongs To

The result is one rung of an infinite ladder. For every odd n and every $k \geq 1$,

$$n^{2^k} \equiv 1 \pmod{2^{k+2}}.$$

The proof is the squaring observation above, applied repeatedly from the base case $n^2 \equiv 1 \pmod{8}$. Verified for $k \leq 6$, and sharp at every stage — each congruence fails at the next power of two:

k	holds	fails
1	$n^2 \equiv 1 \pmod{8}$	$n^2 \not\equiv 1 \pmod{16}$
2	$n^4 \equiv 1 \pmod{16}$	$n^4 \not\equiv 1 \pmod{32}$
3	$n^8 \equiv 1 \pmod{32}$	$n^8 \not\equiv 1 \pmod{64}$

So the problem is the $k = 2$ case, and the base case $n^2 \equiv 1 \pmod{8}$ — the Lemma — is the only one requiring an actual argument.

1.10 Remark — Why the Exponent Is 4 and Not 8

There is a group-theoretic reading that explains the numbers. The odd residues mod **16** form a group under multiplication, the units $(\mathbb{Z}/16\mathbb{Z})^\times$, of order $\varphi(16) = 8$. Euler's theorem therefore gives

$$n^8 \equiv 1 \pmod{16} \quad \text{for odd } n,$$

which is true but weaker than what this problem asserts.

The improvement from 8 to 4 comes from the group's *shape*. Computing the order of each unit gives

n	1	3	5	7	9	11	13	15
order	1	4	4	2	2	4	4	2

so the **exponent** of the group — the least common multiple of the orders — is 4, not 8. The group is $\mathbb{Z}/2 \times \mathbb{Z}/4$ rather than $\mathbb{Z}/8$; it is not cyclic, and no element has order 8.

This is the standard fact that $(\mathbb{Z}/2^m\mathbb{Z})^\times$ fails to be cyclic for $m \geq 3$, unlike $(\mathbb{Z}/p^m\mathbb{Z})^\times$ for odd primes p . Powers of two are the exception, and this exercise is a concrete symptom of it: the exponent is half the order, so the guaranteed power is 4 rather than 8, and the table above shows 4 is attained.